

Introduction to Science, History and branches of Science, Chemical and physical properties of the materials (i.e., solid, liquid, gases), chemical bonding, types of chemical reaction

I. Definition and Scope of Science

– Science: systematic study of the natural and physical world through observation, experimentation, and evidence-based reasoning

– **Branches of science.**

- Natural sciences (biology, chemistry, physics, earth sciences)
- Formal sciences (mathematics, logic)
- Social sciences (psychology, sociology, anthropology)
- Applied sciences (engineering, medicine, technology)

II. Scientific Method

– Steps:

1. Observation
2. Question
3. Hypothesis
4. Experimentation
5. Data analysis
6. Conclusion

– Key principles:

- Empiricism (reliance on evidence)
- Objectivity (neutral perspective)
- Testability (falsifiability)

III. Types of Scientific Inquiry

- Descriptive science (description of phenomena)
- Exploratory science (investigation of unknown phenomena)
- Explanatory science (identification of causes and mechanisms)
- Predictive science (forecasting outcomes)

IV. Scientific Tools and Techniques

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- Measurement and data collection
- Observation and experimentation
- Statistical analysis and modeling
- Laboratory equipment and technology

V. Scientific Theories and Laws

- Theories: well-substantiated explanations for phenomena
- Laws: mathematical descriptions of regularities in nature
- Examples: gravity, evolution, atomic theory

VI. Importance of Science in Society

- Improving quality of life
- Addressing global challenges (climate change, health, sustainability)
- Informing policy and decision-making
- Advancing technology and innovation

VII. Challenges and Limitations of Science

- Uncertainty and error
- Ethical considerations
- Funding and resource constraints
- Interdisciplinary boundaries

Physical and Chemical Properties of Gases:

Physical Properties:

1. Pressure (P): force per unit area.
2. Temperature (T): measured in Kelvin (K).
3. Volume (V): occupied space.
4. Density (ρ): mass per unit volume.
5. Viscosity (μ): resistance to flow.
6. Diffusivity (D): rate of diffusion.

Chemical Properties:

1. Reactivity: ability to undergo chemical reactions.
2. Flammability: ability to burn.

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3. Toxicity: harmful effects on living organisms.
4. Corrosivity: ability to damage materials.
5. Solubility: ability to dissolve in liquids.

Specific Properties of Gases:

1. Boiling Point (BP): temperature at which vapor pressure equals atmospheric pressure.
2. Melting Point (MP): temperature at which solid and liquid phases coexist.
3. Critical Temperature (T_c): temperature above which gas cannot be liquefied.
4. Critical Pressure (P_c): pressure required to liquefy gas at T_c .
5. Gas Constant (R): specific to each gas.

Gas Laws:

1. Boyle's Law ($P_1V_1 = P_2V_2$).
2. Charles' Law ($V_1/T_1 = V_2/T_2$).
3. Avogadro's Law ($V \propto n$).
4. Ideal Gas Law ($PV = nRT$).
5. Real Gas Law (accounting for non-ideal behavior).

Types of Gases:

1. Noble Gases (e.g., He, Ne, Ar).
2. Reactive Gases (e.g., O_2 , Cl_2).
3. Inert Gases (e.g., N_2 , CO_2).
4. Toxic Gases (e.g., HCN, CO).
5. Greenhouse Gases (e.g., CH_4 , N_2O).

Applications:

1. Industrial processes (e.g., manufacturing).
2. Energy production (e.g., power plants).
3. Transportation (e.g., fuel).
4. Medical applications (e.g., anesthesia).
5. Environmental monitoring.

Notable Scientists:

1. Robert Boyle.

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2. Jacques Charles.
3. Amedeo Avogadro.
4. Joseph Gay-Lussac.
5. Svante Arrhenius.

Physical and Chemical Properties of Liquids.

Physical Properties.

1. Density (ρ): mass per unit volume.
2. Viscosity (μ): resistance to flow.
3. Surface Tension (γ): energy per unit area.
4. Boiling Point (BP): temperature at which vapor pressure equals atmospheric pressure.
5. Melting Point (MP): temperature at which solid and liquid phases coexist.
6. Specific Heat Capacity (c): energy required to change temperature.
7. Thermal Conductivity (k): ability to conduct heat.

Chemical Properties.

1. Solubility: ability to dissolve in other liquids.
2. Reactivity: ability to undergo chemical reactions.
3. pH: measure of acidity or basicity.
4. Polarity: ability to form dipoles.
5. Volatility: tendency to evaporate.

Specific Properties of Liquids.

1. Dielectric Constant (ϵ): measure of electrical insulation.
2. Refractive Index (n): bending of light.
3. Freezing Point Depression: decrease in freezing point due to dissolved substances.
4. Boiling Point Elevation: increase in boiling point due to dissolved substances.

Liquid Types.

1. Polar Liquids (e.g., water, ammonia).
2. Nonpolar Liquids (e.g., hexane, benzene).
3. Ionic Liquids (e.g., molten salts).
4. Viscoelastic Liquids (e.g., polymers).

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5. Supercritical Fluids (e.g., high-pressure CO₂).

Applications.

1. Industrial processes (e.g., manufacturing).
2. Pharmaceutical applications (e.g., drug delivery).
3. Energy production (e.g., fuel).
4. Environmental remediation (e.g., pollution cleanup).
5. Food processing (e.g., pasteurization).

Notable Scientists.

1. Antoine Lavoisier.
2. Joseph Gay-Lussac.
3. Michael Faraday.
4. Svante Arrhenius.
5. Irving Langmuir.

Physical and Chemical Properties of Solids.

Physical Properties.

1. Density (ρ): mass per unit volume.
2. Melting Point (MP): temperature at which solid becomes liquid.
3. Boiling Point (BP): temperature at which liquid becomes gas.
4. Hardness: resistance to scratching and abrasion.
5. Thermal Conductivity (k): ability to conduct heat.
6. Electrical Conductivity (σ): ability to conduct electricity.
7. Optical Properties: transparency, refractive index, reflectivity.

Chemical Properties.

1. Reactivity: ability to undergo chemical reactions.
2. Corrosion Resistance: resistance to chemical degradation.
3. Solubility: ability to dissolve in liquids.
4. Chemical Stability: resistance to chemical change.
5. pH: measure of acidity or basicity.

Specific Properties of Solids.

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1. Crystal Structure: arrangement of atoms or molecules.
2. Lattice Energy: energy holding solid together.
3. Surface Energy: energy at the surface of the solid.
4. Pore Size and Distribution: void spaces within the solid.

Types of Solids:

1. Crystalline Solids (e.g., salts, metals).
2. Amorphous Solids (e.g., glasses, polymers).
3. Semi-Crystalline Solids (e.g., fibers, nanomaterials).
4. Porous Solids (e.g., zeolites, activated carbon).

Applications:

1. Construction Materials (e.g., steel, concrete).
2. Electronic Components (e.g., semiconductors, wires).
3. Pharmaceutical Applications (e.g., tablets, capsules).
4. Energy Storage (e.g., batteries, fuel cells).
5. Biomedical Applications (e.g., implants, surgical instruments).

Notable Scientists:

1. Daniel Bernoulli.
2. Antoine Lavoisier.
3. Joseph Gay-Lussac.
4. Dmitri Mendeleev.
5. Linus Pauling.

Chemical Bonding:

Types of Chemical Bonds:

1. Ionic Bonds: electrostatic attraction between oppositely charged ions.
2. Covalent Bonds: sharing of electrons between atoms.
3. Metallic Bonds: delocalization of electrons among metal atoms.
4. Hydrogen Bonds: weak electrostatic attraction between hydrogen and electronegative atoms.
5. Van der Waals Bonds: weak intermolecular forces.

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Chemical Bonding Theories:

1. Valence Shell Electron Pair Repulsion (VSEPR) Theory.
2. Molecular Orbital (MO) Theory.
3. Lewis Structure.
4. Octet Rule.

Factors Affecting Chemical Bonding:

1. Electronegativity.
2. Ionization Energy.
3. Electron Affinity.
4. Atomic Radius.
5. Molecular Geometry.

Properties of Chemical Bonds:

1. Bond Length.
2. Bond Strength.
3. Bond Energy.
4. Bond Order.
5. Bond Polarity.

Applications of Chemical Bonding:

1. Materials Science.
2. Pharmaceutical Industry.
3. Biotechnology.
4. Energy Storage.
5. Catalysis.

Notable Scientists:

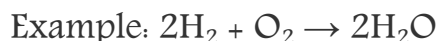
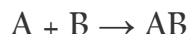
1. Gilbert N. Lewis.
2. Linus Pauling.
3. Erwin Schrödinger.
4. Walter Heitler.
5. Fritz London.

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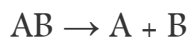
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Types of Chemical Reactions:

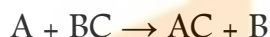
1. Synthesis Reactions (Combination)



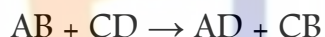
2. Decomposition Reactions



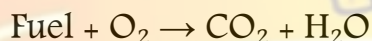
3. Replacement Reactions (Single Displacement)



4. Double Displacement Reactions



5. Combustion Reactions



6. Oxidation-Reduction (Redox) Reactions

Oxidation: loss of electrons

Reduction: gain of electrons

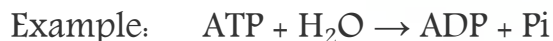


7. Acid-Base Reactions (Neutralization)



8. Hydrolysis Reactions

Water breaks a bond



9. Condensation Reactions

Two molecules join, releasing water

Example: amino acids $\rightarrow$ protein

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10. Polymerization Reactions

Monomers form polymers

Example: ethene $\rightarrow$ polyethylene

Other Types:

1. Isomerization Reactions
2. Elimination Reactions
3. Addition Reactions
4. Electrophilic Substitution Reactions
5. Nuclear Reactions

Reaction Classifications:

1. Exothermic (releases heat)
2. Endothermic (absorbs heat)
3. Reversible
4. Irreversible
5. Catalyzed

Notable Scientists:

1. Antoine Lavoisier
2. Joseph Priestley
3. Humphry Davy
4. Svante Arrhenius
5. Linus Pauling

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